

# Ibrahim Fadhil Djauhari

✉ [Email](#) | 🌐 [Website](#) | 📄 [Google Scholar](#) | [in](#) [LinkedIn](#) | [GitHub](#)

## Education

---

**Master of Science in Marine Engineering & Autonomy** at the University of Southampton, 2023 – 2024

**Bachelor of Science in Electrical Engineering** at Bandung Institute of Technology (ITB), 2016 – 2020

## Work Experience

---

### Robotics Engineering Lead

eBdesk Teknologi Group, November 2025 – present

- Conducted technical pre-sales demonstrations of robotic capabilities to five prospective clients
- Evaluated the feasibility of building OEM mobile robot, aerial drone, and RF products during a 5-day vendor assessment in China
- Trained a YOLO-based computer vision model for object detection and optimized it to run with >30fps on an Nvidia Jetson Orin Nano
- Designed and implemented a quadrupedal robot SLAM system using ROS2 implemented on an Nvidia Jetson Orin Nano. The system reads 5Hz 3D point cloud data from a LIDAR sensor and outputs 2D map for robot mapping and localization
- Designed and implemented a multimodal robot agent, integrating hybrid cloud and edge LLMs to interpret user voice commands and real-time sensor readings to command robots with minimal latency. Validated in office-scale testing on three quadrupedal and wheeled robots with diverse sensor and actuator configurations
- Designed and implemented an IoT-based greenhouse monitoring and control system, achieving 100% monitoring uptime of four environmental variables (Temperature, humidity, light intensity, and color spectrum) and controlling three actuators (Lights, fan, and humidifier) to ensure healthy growth of Anthurium plants
- Founded the company's robotics engineering team and lead a multidisciplinary team of Robotics, IoT, AI, and Electrical Engineers

### Underwater Drone Operator & Marketing

Rovostech Co. Ltd., August 2025 – November 2025

- Performed maintenance and assembly of electrical and mechanical hardware for custom-built Remotely Operated Vehicles (ROVs)
- Planned and performed a three-day shipwreck mapping mission on the Java Sea using a 150m-rated observation-class ROV, successfully gathering over 10,000 images of underwater objects from multiple angles
- Built a CUDA-based photogrammetry pipeline integrating computer vision for image correction and object detection, successfully converting a ~10,000 image dataset into two high-fidelity 3D models
- Presented and advised ROV technical specifications to align with the project requirements of potential clients

### Researcher in the Ocean Perception Lab

University of Southampton, September 2024 – April 2025

- Developed a novel vision-based motion planning system for autonomous tracking of subsea cables using C++, Python, and ROS, resulting in two academic publications.
- Trained and implemented a real-time computer vision/perception model on an Nvidia Jetson board using Pytorch, achieving a 1 second cable detection rate in 4k resolution seafloor images
- Developed a graph-based SLAM system to update map estimations based on new visual observations in real-time, successfully proving robustness in up to 2% false positive and 25% false negative visual observation rates
- Planned and performed eight days of field-testing of the subsea cable tracking system on an Autonomous Underwater Vehicle (AUV), successfully observing 60% of a 110m subsea cable autonomously

### Cloud Middleware Engineer

Samsung R&D Institute Indonesia, August 2021 – August 2023

- Developed and deployed a Point-of-Interest (POI) search application serving over 100 million monthly requests for Samsung services in the Europe & Asia Pacific regions using Amazon Web Services (AWS), ensuring 100% uptime using Blue-Green deployment strategy

### Research Assistant in the Computer & Control Systems Lab (LSKK ITB)

Bandung Institute of Technology (ITB), October 2020 – August 2023

- Developed a line-of-sight guidance system and a gain-scheduled PID control system for precise motion planning of an AUV, achieving a maximum of 6m cross-track error in a 2500m<sup>2</sup> lawnmower survey pattern
- Developed a navigation system integrating multiple sensors (IMU, DVL, Altimeter, GPS) of an AUV fused using an Extended Kalman Filter (EKF), achieving International Hydrographic Organization (IHO) error standards for hydrographic surveys
- Performed Hardware-In-The-Loop-Simulations (HILS) to test system performance on embedded AUV hardware using Rviz and Gazebo

## Leadership Experience

---

### Project Leader

OSKM ITB 2019, April 2019 – September 2019

- Managed 2,481 committee members and over 2 billion IDR of funds to achieve the vision of [OSKM ITB 2019](#) as a student orientation event to welcome and educate 4,465 new ITB students
- Helped raise over 600 million IDR in funding by negotiating deals with sponsors such as Telkom Indonesia, Paragon, and Ultramilk
- Initiated OSKM ITB 2019 as the largest zero-waste event in ITB and recycled 412kg of food waste throughout the event

## Publications & Patents

---

**I. F. Djauhari**, A. Bodenmann, S. Simmons, C. Liang, T. Bennetts, D. Newborough, and B. Thornton, "Autonomous Subsea Cable Search and Tracking using Graph-based Optimisation based on Uncertain Priors and Real-time Visual Servoing," *IEEE Journal of Oceanic Engineering Special Issue AUV 2026*, submitted (2025).

**I. F. Djauhari**, A. Bodenmann, S. Simmons and B. Thornton, "[Subsea Cable Search and Path Estimation Using Graph SLAM for AUV-Based Inspection](#)," *2025 IEEE Underwater Technology (UT)*, Taipei, Taiwan, 2025, pp. 1-6, doi: 10.1109/UT61067.2025.10947380.

A. Bodenmann, **I. F. Djauhari**, J. Devgon and B. Thornton, "[Real-time Subsea Communication Cable Detection for AUV-based Inspection](#)," *2024 IEEE/OES Autonomous Underwater Vehicles Symposium (AUV)*, Boston, MA, USA, 2024, pp. 1-6, doi: 10.1109/AUV61864.2024.11030782.

"*Alat dan Metode Untuk Menggerakkan Hybrid Autonomous Underwater Glider dan Metode untuk Memandu Gerakannya*". Indonesia Patent No. [IDP000101328](#), filed 22 July 2021, issued 11 December 2024.

"*Sistem dan Metode Navigasi untuk Menentukan Data Posisi, Orientasi dan Kecepatan Wahana Luncur Bawah Air Otonom Hibrida*". Indonesia Patent No. [IDP000095871](#), filed 15 June 2021, issued 10 October 2024.

## Training & Workshops

---

**Marine Data 4 Sustainable Oceans Workshop (2025)**: Attended a marine data processing workshop organized by Copernicus Marine Service and Mercator Ocean International

**Sustainathon Indonesia (2024)**: Developed the prototype for "[SejukHemat](#)", an application to help optimize AC temperature for power efficiency, in a six-week ESG-themed virtual hackathon